Earth's Rotation and the Daily Motion of the Sky

What you will be able to do

1. Explain why the Sun, Moon and stars rise in the east and set in the west.
2. Describe how the stars move when you face north, south, east and west.
3. Find your latitude from the altitude of Polaris (*Dhruva Tara*).
4. Explain the difference between a **solar day** (24 h) and a **sidereal day** (23 h 56 m).
5. Calculate how much earlier a star rises after a given number of days.

1. The sky is a giant clock

Watch the sky for just one hour and you will notice that everything moves. The Sun climbs in the morning and sinks in the evening. At night the stars rise in the east, arc across the sky and set in the west. The whole sky seems to turn *as one piece*, like a huge dome spinning above us.

For thousands of years people believed that the dome really was spinning. Today we know it is the other way round: **the Earth spins, and we spin with it.**

Indian connection — Aryabhata (499 CE)

In his book *Aryabhatiya*, Aryabhata of Kusumapura (near modern Patna) wrote that it is the **Earth that rotates**. He compared it to a person in a moving boat who sees the trees on the shore moving backwards. In the same way, the "fixed" stars appear to move westward because we are carried eastward. This idea was more than a thousand years ahead of its time in Europe.

2. How fast does the Earth spin?

- Earth turns once on its axis from **west to east**.
- Seen from above the North Pole, it turns **anticlockwise**.
- One turn relative to the distant stars takes **23 h 56 m 4 s** (a *sidereal day*).

Because the sky turns through 360° in about 24 hours:

$$\frac{360^\circ}{24\ \text{h}} = 15^\circ \text{ per hour} \quad \Longrightarrow \quad 1^\circ \text{ every 4 minutes}, \qquad 15' \text{ per minute}.$$

The Sun and the Moon are each about $\frac{1}{2}^\circ$ wide, so the sky turns by "one Sun-width" in about **2 minutes**. That is roughly how long the Sun takes to sink below a flat horizon.

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P.T.O.

Speed of a point on the Earth

Earth's equator is about 40,075 km long and one turn takes about 24 h, so a person on the equator moves at

$$v_{\text{eq}} \approx \frac{40{,}075\ \text{km}}{24\ \text{h}} \approx 1{,}670\ \text{km/h}.$$

Away from the equator you travel on a smaller circle of radius $R\cos\varphi$, where φ is the latitude:

$$\boxed{v \approx 1670\cos\varphi\ \ \text{km/h}}$$

Worked example

Mumbai is at latitude $\varphi = 19^\circ$ N.

$$v = 1670 \times \cos 19^\circ = 1670 \times 0.946 \approx 1{,}580\ \text{km/h}.$$

That is faster than a passenger jet — yet you feel nothing, because everything around you moves with you.

3. Watching the sky turn



A real long-exposure photograph: every star draws an arc around one point — the north celestial pole, close to Polaris. It was taken at latitude 19.8° N, almost the same as Mumbai, so the pole is about 20° above the horizon, just as from central India.

Photo: International Gemini Observatory/NOIRLab/NSF/AURA/T. Matsopoulos, CC BY 4.0, via Wikimedia Commons

What you see depends on which way you face:

Facing	What the stars do
North	Circle **anticlockwise** around Polaris. Stars close to Polaris never set — they are *circumpolar*.

(continued)

Facing	What the stars do
East	Rise up out of the horizon, climbing at a slant towards the south.
South	Move from left (east) to right (west) in low arcs, reaching their highest point due south.
West	Sink into the horizon at a slant.

How steeply do stars rise?

At latitude φ, stars rise at an angle of $\boxed{90^\circ - \varphi}$ to the horizon.

- At the equator ($\varphi = 0^\circ$) stars rise **straight up**.
- In Mumbai ($\varphi \approx 19^\circ$) they rise at 71° — very steep. That is why twilight is short in India.
- At the North Pole ($\varphi = 90^\circ$) stars never rise or set; they circle parallel to the horizon.

4. Polaris, the "fixed" star (Dhruva Tara)

The Earth's axis points almost exactly at a moderately bright star called **Polaris** (the Pole Star, **Dhruva**). Because it sits almost on the axis, Polaris barely moves while every other star circles around it.

Indian connection — the story of Dhruva

In the Puranas, the child Dhruva meditated with such unshakable determination that he was given a place in the heavens that never moves. Polaris has been the symbol of steadiness ever since. (Strictly, Polaris is about 0.7° from the true pole, so it traces a tiny circle each day.)

How to find Polaris

Look for seven bright stars shaped like a ladle or a question mark in the northern sky — the **Saptarishi** (Big Dipper). The two stars at the outer edge of its "bowl" are called the *pointers*. Draw a line from the lower pointer through the upper one and extend it about five times their separation: you will reach Polaris, a lone star of moderate brightness. When the Saptarishi is below the horizon (October–December evenings), look instead for a bright "W" of five stars low in the north; Polaris lies on the side that the W opens towards.

The most useful rule in naked-eye astronomy

Altitude of Polaris above the northern horizon $\approx$ your latitude φ

Place	Latitude	Height of Polaris
Srinagar	34.1° N	34° — high and easy
New Delhi	28.6° N	29°
Kolkata	22.6° N	23°
Mumbai	19.1° N	19°
Bengaluru / Chennai	13° N	13° — low

(continued)

Place	Latitude	Height of Polaris
Thiruvananthapuram	8.5° N	8.5° — very low, often lost in haze
Equator	0°	on the horizon

Your hand is a protractor

Hold your hand at **full arm's length**. People with longer arms tend to have bigger hands, so this works for almost everyone.

Hand at arm's length	Angle
Width of the little finger	≈ 1°
Three middle fingers together	≈ 5°
Closed fist (across the knuckles)	≈ 10°
Spread hand, thumb tip to little-finger tip	≈ 20°–25°
Horizon to the point overhead	90° (= 9 fists)

5. The solar day and the sidereal day

If Earth turns once in 23 h 56 m, why is a day 24 hours long?

The Earth goes once around the Sun (360°) in about 365 days, i.e. about 1° per day. Turning that extra 1° takes

$$1^\circ \times 4 \text{ min per degree} \approx 4 \text{ min}.$$

	Measured relative to	Length
Sidereal day	distant stars	**23 h 56 m 4 s**
Solar day	the Sun	**24 h 00 m** (average)
Difference		≈ 3 m 56 s ≈ **4 minutes**

What this means for stargazers

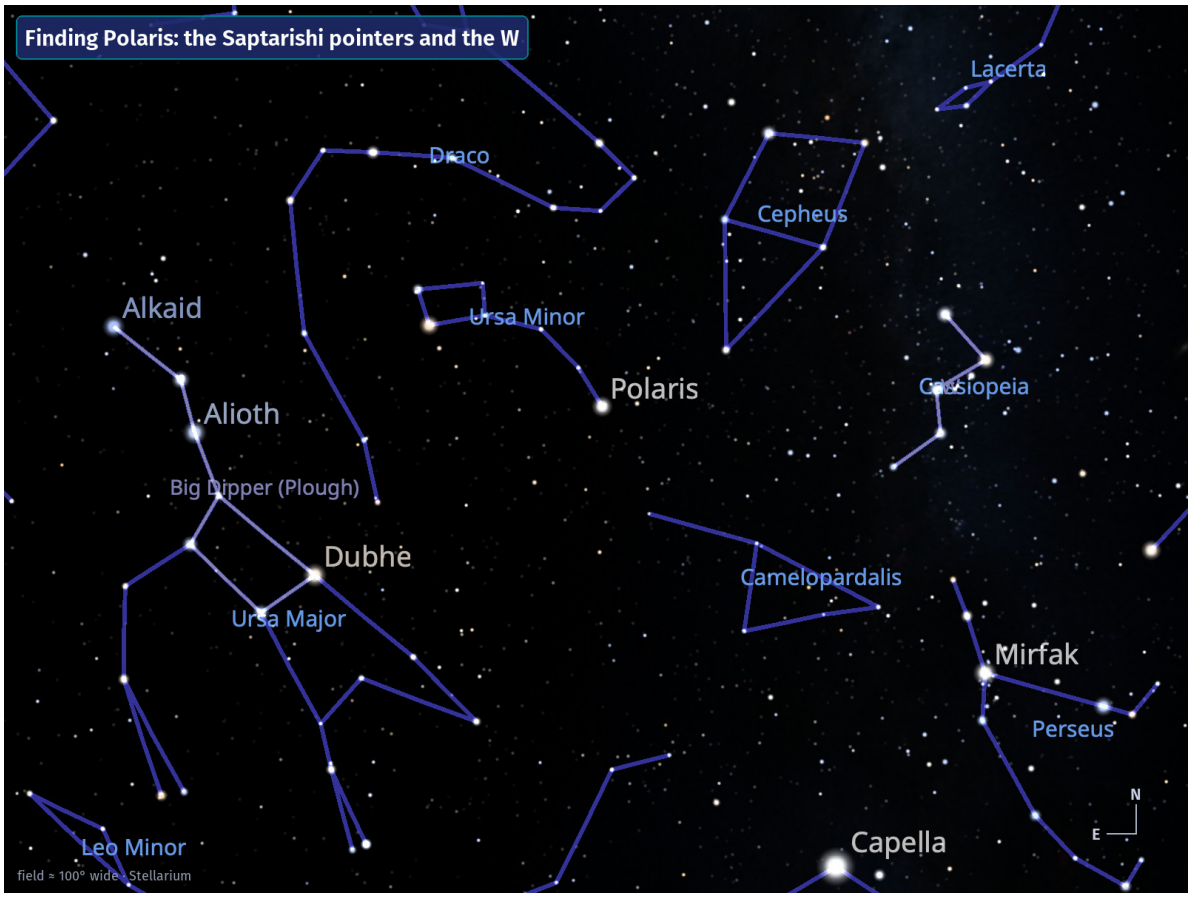
- Each star rises about **4 minutes earlier** every night.
- After 15 nights that adds up to 1 hour, and after a month to about **2 hours**.
- After 12 months it adds up to 24 hours — and the sky is back where it started.

This is why **different constellations are seen in different seasons**: the star patterns that rule the evening sky in December are not the ones that rule it in June.

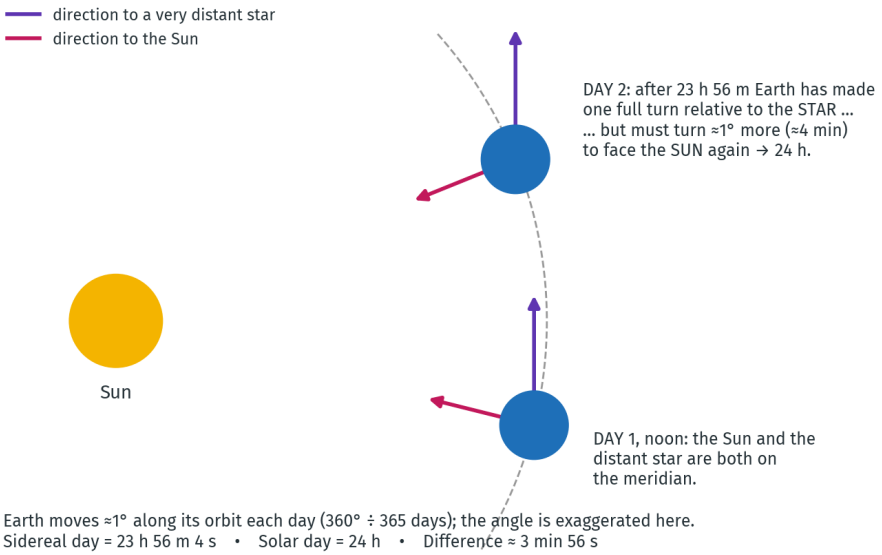
Worked example

On 1 December the star Sirius rises at about 10:00 PM. When will it rise on 1 January, 31 days later?

$$\Delta t = 31 \times 3.93 \text{ min} \approx 122 \text{ min} = 2 \text{ h } 2 \text{ min (earlier)}$$



The northern sky as drawn by Stellarium. Extend the line through the two pointer stars of the Saptarishi (Merak → Dubhe) about five times to reach Polaris; the "W" (Cassiopeia) lies on the other side of Polaris.



While Earth spins, it also moves about 1° along its orbit each day, so it must turn about 1° extra to bring the Sun back to the same place in the sky.

Worked example *(continued)*

Rising time $\approx$ 10:00 PM $-$ 2:02 $=$ **about 7:58 PM**.

6. Proof that the Earth really turns

The stars' daily motion could, in principle, be explained by a spinning sky. How do we know it is the Earth that spins?

1. **Foucault's pendulum (1851).** A long, heavy pendulum keeps swinging in a fixed plane while the Earth turns underneath it, so its swing slowly appears to rotate. The time for one full rotation is

$$T = \frac{23.93\ \text{h}}{\sin\varphi}.$$

   In Mumbai: $T = 23.93/\sin 19^\circ = 23.93/0.326 \approx 73.5\,\text{h}$. India's new Parliament building in New Delhi has a Foucault pendulum, made by the National Council of Science Museums.
2. **Weather.** Cyclones in the northern hemisphere spin anticlockwise because of Earth's rotation (the Coriolis effect).
3. **Earth's shape.** Earth is about 43 km wider across the equator than through the poles — just what a spinning ball should be.
4. **Space photographs.** Time-lapse images from satellites show the Earth turning.



A Foucault pendulum in a science museum. As the floor (the Earth) turns, the swinging bob knocks over the pegs one by one.

Photo: Stephen Rahn, CC0, via Wikimedia Commons

Try tonight

1. **Find north.** Stand where you can see the northern horizon (use a compass app if needed).
2. **Find Polaris** using the Saptarishi pointers or the "W", as described above. In India it is low in the north and not very bright.
3. **Measure its height** with your fist at arm's length. Does it roughly match your city's latitude?
4. **Watch a bright star near the eastern horizon** for 30 minutes against a fixed object such as a tree or a building. How far has it moved? (Expected: $30 \times 15' = 7.5°$, about one fist.)

Key points to remember

- Earth rotates **west → east**, so the sky appears to turn **east → west**.
- The sky turns 15° per hour, i.e. 1° every 4 minutes.
- **Altitude of Polaris ≈ your latitude.** Fist ≈ 10°, little finger ≈ 1°.
- Sidereal day = 23 h 56 m 4 s; solar day = 24 h. Stars rise ~4 min earlier each night (2 h per month).
- Aryabhata proposed Earth's rotation in 499 CE.

Check yourself

1. If you face south at night, do stars move from left to right or right to left?
2. A friend in Delhi and a friend in Chennai both look for Polaris. Who sees it higher, and by roughly how much?
3. A star pattern is overhead at midnight tonight. Roughly when will it be overhead three months from now?
4. Why does twilight last longer in Europe than in India?

Answers: 1. Left to right (east to west). 2. Delhi, by about 15°. 3. Around 6 PM (3×2 h = 6 h earlier). 4. At high latitude the Sun sinks at a shallow slant, so it takes longer to get 18° below the horizon.

Further reading (free / public)

- OpenStax, *Astronomy 2e* (free online textbook), §2.1 "The Sky Above" and §4.1 "Earth and Sky" — `https://openstax.org/books/astronomy-2e/pages/2-1-the-sky-above`
- **Stellarium** (free planetarium software, `https://stellarium.org`): set your city and speed up time to watch the sky turn.
- H. A. Rey, *The Stars: A New Way to See Them* — a classic, beautifully illustrated beginner's book.